

Limits Graphically and Numerically

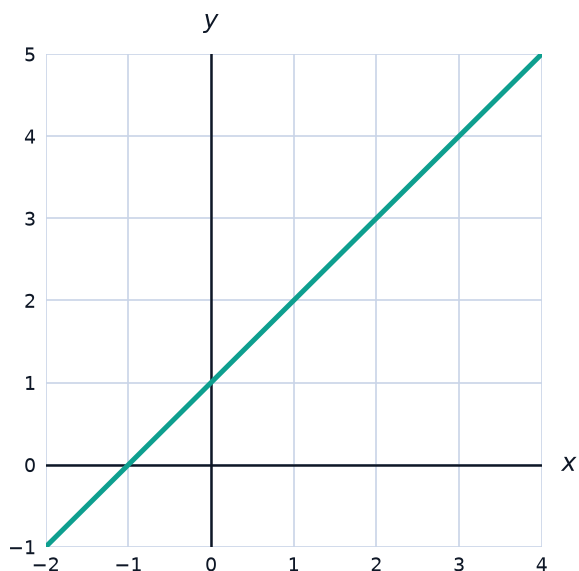


Estimating limits from a table

- 1 A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \rightarrow 3} g(x)$.
- 2 Using a table of values with x near 0 (radians), estimate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Reading limits from a graph

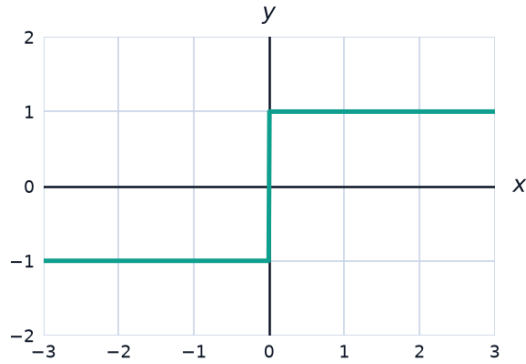
- 3 The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ ($x \neq 1$), state $\lim_{x \rightarrow 1} f(x)$.



- 4 A function h has $\lim_{x \rightarrow 2^-} h(x) = 5$ and $\lim_{x \rightarrow 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \rightarrow 2} h(x)$ and explain whether h is continuous at $x = 2$.

One-sided limits and limits that fail to exist

- 5 For $f(x) = \frac{x}{|x|}$, find:



- a $\lim_{x \rightarrow 0^+} f(x)$ b $\lim_{x \rightarrow 0^-} f(x)$ c $\lim_{x \rightarrow 0} f(x)$
- 6 A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and state whether $\lim_{x \rightarrow 1} f(x)$ exists.

Algebraic evaluation

- 7 Evaluate each limit by direct substitution or factoring:

a $\lim_{x \rightarrow 4} (3x^2 - 5x + 1)$

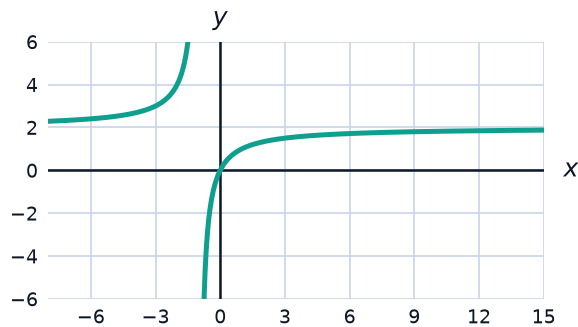
b $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

c $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$

- 8 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$ by rationalizing the numerator.

Limits at infinity from a graph

- 9 The graph shows $f(x) = \frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x \rightarrow \infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x .



Limits involving infinite discontinuities

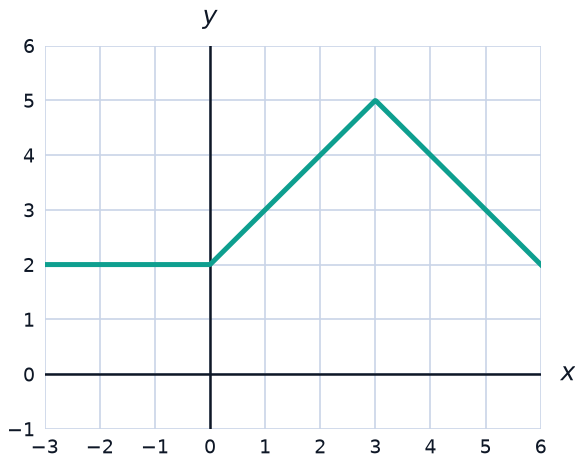
- 10 For $f(x) = \frac{1}{x-3}$, find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.
- 11 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2+1}{x^2-4}$, using the fact that the numerator and denominator have the same degree.

Estimating limits with a calculator-style table

- 12 Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1, -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

Limits with piecewise graphical behavior

- 13 A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \leq x < 3$, and $f(x) = 8 - x$ for $x \geq 3$. Find $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 3} f(x)$, stating whether each exists.



- 14 Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

Multiple choice — easy

- 15 Find $\lim_{x \rightarrow -3} (x^2 + 3x - 4)$.
- a. -3
 - b. -4
 - c. 4
 - d. -5

16 Find $\lim_{x \rightarrow 4} (5x^2 + 5x + 9)$.

a. -109

b. 108

c. 109

d. 110

17 Find $\lim_{x \rightarrow 3} (4x^2 - x + 8)$.

a. 41

b. 42

c. 40

d. -41

18 Find $\lim_{x \rightarrow 3} (5x^2 + 3x - 5)$.

a. 48

b. 50

c. -49

d. 49

19 Find $\lim_{x \rightarrow 3} (5x^2 + 7)$.

a. -52

b. 51

c. 52

d. 53

20 Find $\lim_{x \rightarrow 1} (x^2 + 2x - 9)$.

a. -6

b. -7

c. -5

d. 6

21 Find $\lim_{x \rightarrow 4} (5x^2 - 8x + 5)$.

a. 54

b. 53

c. -53

d. 52

22 Find $\lim_{x \rightarrow -1} (5x^2 - 6x - 8)$.

a. 3

b. 4

c. -3

d. 2

23 Find $\lim_{x \rightarrow -3} (2x^2 - 9x + 8)$.

a. 52

b. -53

c. 54

d. 53

24 Find $\lim_{x \rightarrow -2} (x^2 - 4x + 6)$.

a. 17

b. 19

c. -18

d. 18

25 Find $\lim_{x \rightarrow 4} (4x^2 + 3x + 2)$.

a. 77

b. 78

c. 79

d. -78

26 Find $\lim_{x \rightarrow 3} (4x^2 + 3x - 5)$.

a. 39

b. 40

c. 41

d. -40

27 Find $\lim_{x \rightarrow -4} (3x^2 - 6x - 4)$.

a. -68

b. 69

c. 67

d. 68

28 Find $\lim_{x \rightarrow 1} (2x^2 - x - 3)$.

- a. 2
- b. -3
- c. -2
- d. -1

29 Find $\lim_{x \rightarrow 0} (4x^2 + 9x - 5)$.

- a. -6
- b. -4
- c. -5
- d. 5

Multiple choice — medium

30 Evaluate $\lim_{x \rightarrow 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

- a. -3
- b. 1
- c. -1
- d. 3

31 Evaluate $\lim_{x \rightarrow -2} \frac{(x+2)(x-5)}{x+2}$ by factoring and cancelling.

- a. 7
- b. -7
- c. -5
- d. -9

32 Evaluate $\lim_{x \rightarrow 2} \frac{(x-2)(x-5)}{x-2}$ by factoring and cancelling.

- a. 3
- b. -1
- c. -5
- d. -3

33 Evaluate $\lim_{x \rightarrow 5} \frac{(x-5)(x-6)}{x-5}$ by factoring and cancelling.

- a. -1
- b. 1
- c. 11
- d. -3

34 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x+3)}{x+4}$ by factoring and cancelling.

- a. -3
- b. -1
- c. -7
- d. 1

35 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x-5)}{x+4}$ by factoring and cancelling.

- a. -11
- b. -7
- c. -9
- d. 9

36 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x+5)}{x+4}$ by factoring and cancelling.

- a. 3
- b. 1
- c. -9
- d. -1

37 Evaluate $\lim_{x \rightarrow -1} \frac{(x+1)(x-6)}{x+1}$ by factoring and cancelling.

- a. -9
- b. -5
- c. 7
- d. -7

38 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x+4)}{x-6}$ by factoring and cancelling.

- a. -10
- b. 8
- c. 10
- d. 12

39 Evaluate $\lim_{x \rightarrow -5} \frac{(x+5)(x-3)}{x+5}$ by factoring and cancelling.

- a. -8
- b. -6
- c. 8
- d. -10

40 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x-3)}{x-6}$ by factoring and cancelling.

- a. 1
- b. -3
- c. 3
- d. 5

41 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x+3)}{x-6}$ by factoring and cancelling.

- a. 11
- b. 9
- c. 7
- d. -9

42 Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$ by factoring and cancelling.

- a. 4
- b. -6
- c. 8
- d. 6

43 Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$ by factoring and cancelling.

- a. -4
- b. 4
- c. 6
- d. 2

- 44 Evaluate $\lim_{x \rightarrow -5} \frac{(x+5)(x+1)}{x+5}$ by factoring and cancelling.
- a. -4
 - b. -6
 - c. -2
 - d. 4
-

Multiple choice — hard

- 45 For $f(x) = \frac{-1}{x+5}$, find $\lim_{x \rightarrow -5^-} f(x)$.

- a. -5
- b. $+\infty$
- c. 0
- d. $-\infty$

- 46 Evaluate $\lim_{x \rightarrow \infty} \frac{6x^2 + 3x}{8x^2 + 8x}$.

- a. $\frac{4}{3}$
- b. $\frac{3}{4}$
- c. $\frac{7}{8}$
- d. $\frac{2}{3}$

- 47 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 4x}{2x^2 - 6x}$.

- a. 2
- b. $\frac{1}{2}$
- c. $\frac{5}{2}$
- d. $\frac{4}{3}$

48 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^3 + 2x}{6x^2 + 1}$.

- a. 0
- b. 1
- c. does not exist ($+\infty$)
- d. $\frac{7}{6}$

49 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 + 6x}{9x^2 - 3x}$.

- a. $\frac{5}{9}$
- b. $\frac{1}{2}$
- c. $\frac{9}{5}$
- d. $\frac{2}{3}$

50 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^2 - 9x}{2x^2 - 6x}$.

- a. $\frac{2}{7}$
- b. 4
- c. $\frac{7}{2}$
- d. $\frac{7}{3}$

51 Evaluate $\lim_{x \rightarrow \infty} \frac{6x + 3}{7x^2 - 1}$.

- a. does not exist
- b. 0
- c. $\frac{6}{7}$
- d. 1

52 For $f(x) = \frac{1}{x+5}$, find $\lim_{x \rightarrow -5^-} f(x)$.

- a. 0
- b. $+\infty$
- c. -5
- d. $-\infty$

53 For $f(x) = \frac{-1}{x-2}$, find $\lim_{x \rightarrow 2^-} f(x)$.

- a. 2
- b. 0
- c. $-\infty$
- d. $+\infty$

54 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^2 - 7x}{4x^2 + 6x}$.

- a. 2
- b. $\frac{9}{4}$
- c. $\frac{1}{2}$
- d. $\frac{8}{5}$

55 For $f(x) = \frac{-1}{x+6}$, find $\lim_{x \rightarrow -6^-} f(x)$.

- a. -6
- b. 0
- c. $+\infty$
- d. $-\infty$

56 Evaluate $\lim_{x \rightarrow \infty} \frac{8x+3}{x^2-1}$.

a. does not exist

b. 8

c. 0

d. 1

57 Evaluate $\lim_{x \rightarrow \infty} \frac{9x^3+2x}{2x^2+1}$.

a. does not exist ($+\infty$)

b. 1

c. $\frac{9}{2}$

d. 0

58 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^3+2x}{2x^2+1}$.

a. 0

b. 1

c. does not exist ($+\infty$)

d. 4

59 Evaluate $\lim_{x \rightarrow \infty} \frac{x^3+2x}{5x^2+1}$.

a. does not exist ($+\infty$)

b. 1

c. $\frac{1}{5}$

d. 0